$\boxed{\text{Case 8:}}$ $x=2, y=4$ $\therefore x^4+y^4=16+256=272<625$

$\boxed{\text{Case 9:}}$ $x=2, y=5$ $\therefore x^4+y^4 = 16+625=641>625$

$\boxed{\text{Case 10:}}$ $x=3, y=3$ $\therefore x^4+y^4=81+81=162<625$

$\boxed{\text{Case 11:}}$ $x=3, y=4$ $\therefore x^4+y^4=81+256=337<625$

$\boxed{\text{Case 12:}}$ $x=4, y=4$ $\therefore x^4+y^4=256+256=512<625$

$\therefore$ There are no solutions to the positive integers x and y for the equation $\boxed{x^4+y^4=625}$ □

34. Prove that there are infinitely many solutions in positive integers x, y and z to the equation $x^2+y^2=z^2$.

$\boxed{\text{Solution:}}$ Suppose, for the sake of contradiction, there are finitely many solutions in positive integers x, y and z to the equation $x^2+y^2=z^2$.

Let p and q be the largest such pair of integers, s.t. $p^2+q^2=r^2$ is True where s and t and r is an integer. ($\forall$ integers s and t, $p\geq s$ and $q\geq t$ where s and t satisfy the given eqn. by substitution as x and y respectively)

$\therefore 4(p^2+q^2)=4r^2 \Rightarrow 4p^2+4q^2=4r^2 \Rightarrow (2p)^2+(2q)^2=(2r)^2$.

$\because p, q, r$ are integers. $\therefore 2p, 2q$ and $2r$ are integers.

$2p>p$, $2q>q$ $\because p, q$ are positive. $\therefore$ Our claim that p and q are the largest such pair of integers s.t. $p^2+q^2=r^2$ is True where r is an integer is False.

$\therefore$ There are infinitely many solutions in positive integers x, y and z to the equation $x^2+y^2=z^2$. □

35. Adapt the proof in Example 4 in Section 1.7 to prove that if $n=abc$, where a, b and c are positive integers, then $a\leq\sqrt[3]{n}$, $b\leq\sqrt[3]{n}$ or $c\leq\sqrt[3]{n}$